Degrade What the Task Ignores: Workload-Conditional Datapaths for State-Space Models on Neuromorphic Hardware

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Abstract

State-space models are often described as the natural sequence architecture for neuromorphic hardware, because their core is a linear recurrence that analog substrates physically implement. But "an SSM on neuromorphic hardware" hides a three-way fork over *where* precision is given up: an exact digital state, a continuous state with a spiking output, or an analog state that is lossy by construction. We compare all three, plus a spiking-state control, at *matched parameters* (identical tensors; only the position of the nonlinearity moves) and at *matched communication rate*, on one statistical workload (character-level language modelling) and one precise-recall workload (synthetic copy at three memory loads), with three seeds per cell.

Neither neuromorphic route dominates, and the ranking *inverts* between the two tasks. On char-LM the analog-state datapath is the best non-digital variant and buys a 1.60× cut in proxy energy for +0.160 bpc (5.3%) of quality; we reach that number by retracting our own intermediate finding that the analog state beat the digital baseline, because giving the baseline the same training-time state noise (σ=0.02) is worth −0.309 bpc — 2.1× the apparent advantage — so the earlier comparison was against an under-regularized comparator. On copy the order reverses: output-spiking retains 0.87 and 0.42 of the regularized baseline's above-chance accuracy margin at M=16 and M=32, the analog state retains 0.50 and 0.28, and a one-bit spiking state sits at *exactly chance* at M=32 and M=64 while being the cheapest cell in the study — an energy number that purchased nothing.

We organize both orderings under one claim: a neuromorphic variant is cheap exactly when it degrades the part of the datapath the task does not depend on. Its sharpest test is out-of-sample and post-hoc: the same 6-bit state quantizer costs −0.002 bpc on char-LM and +0.369/+0.385 bpc on copy at two memory loads. We also report a negative on the theory side. A firing-floor bound from prior work, $\rho \geq H_b^{-1}(M \log_2 K/H)$, is tested by shrinking the recurrent width so the predicted floor swings 12× on a task the reference model still learns; measured spiking-state activity *falls* (0.496 → 0.250) as the floor rises, and accuracy *improves* as predicted capacity shrinks, so the observed activity is width-determined rather than information-determined and the bound is not empirically validated in either direction at this scale. All energy figures are 45 nm operation-count proxies from simulation, not silicon measurements, and the analog route's storage element and converter are unpriced.

1 Introduction

State-space models (SSMs) look like the natural sequence engine for neuromorphic hardware. An SSM's core is a *linear* recurrence, $\dot{x} = Ax + Bu$, $y = Cx$ — a continuous-time dynamical system rather than a stack of nonlinear matrix products. Analog and mixed-signal substrates already *are* linear dynamical systems: a leaky capacitor, or a memristor's short-term relaxation, physically implements e^{At} without anyone having to compute it. The pairing has a working existence proof: the Legendre Memory Unit, itself a linear SSM, has been deployed on both a digital spiking chip and an analog mixed-signal one (Voelker et al., 2019).

But "an SSM on neuromorphic hardware" is not one design. It hides a three-way fork over *where in the datapath precision is given up*:

1. **spike the state** — carry the recurrent state itself in binary events, the canonical spiking-RNN design;
2. **spike the output** — keep the linear state exact and binarize only what leaves the recurrence, the design point of the current spiking-SSM literature (Zhong et al., 2024);
3. **hold the state in analog** — keep the state continuous but accept the imperfections of a physical storage element, and communicate by events, the compute-in-memory route (Zhang et al., 2026).

Each route has papers arguing for it, and each reports energy savings against a baseline of its own choosing. What does not exist is a controlled comparison: the three routes have never been run as the *same* model, at matched parameters, on the same tasks, at matched communication rates. Without that, "spiking SSMs are efficient" and "analog SSMs are efficient" are not comparable statements, and a hardware designer choosing a datapath has no basis for the choice.

This paper supplies that comparison, and its result is that the choice is *workload-conditional*. We build one SSM in four parameter-matched implementations — the three routes above plus the exact digital baseline — in which every variant owns an identical set of tensors and only the *position* of the nonlinearity moves. We run them on a statistical sequence task (character-level language modelling) and a precise-recall task (synthetic copy at three memory loads), at three seeds throughout, and we compare them at matched emitted-event rates so that no variant is credited for merely sitting at a sparser operating point.

The ranking inverts. On character-LM the analog-state route is the best neuromorphic datapath and output-spiking is the worst; on copy the order is exactly reversed, and the spiking-state route does not degrade precise recall so much as *eliminate* it, sitting at exactly chance accuracy at 32 and at 64 symbols of memory load while being the cheapest variant by the energy proxy. Neither non-digital route dominates, so there is no single number of the form "analog retains $X\%$ of digital quality" to quote.

One principle explains both orderings. The obvious candidate explanation — that what matters is how exact the recurrent *state* is — fails: the output-spiking variant has the most exact state of the three on *both* tasks (its measured state activity is exactly 1.0000 in every cell we ran) yet it is best on one task and worst on the other. What does predict both orderings is a claim about the workload rather than the datapath:

> *a neuromorphic SSM datapath is cheap exactly when it degrades the part of the computation the workload does not depend on.*

Statistical sequence modelling needs an exact graded *output* distribution and tolerates a lossy state; precise recall needs an exact *state* and tolerates a binarized output. We test this out of sample with a degradation neither task was designed around: the same 6-bit state quantizer costs -0.002 bpc on char-LM (nothing) and $+0.369/+0.385$ bpc on copy at $M{=}16$ and $M{=}32$. That test is post-hoc identified, and we label it as such throughout (§6.3).

We report our own retraction. An intermediate version of the char-LM result had the analog-state SSM *beating* the digital baseline by 0.149 ± 0.014 bpc. A control showed that finding to be an artifact of an under-regularized comparator: giving the digital baseline the same training-time state noise the analog datapath supplies for free is worth -0.309 bpc, which is $2.1\times$ the entire apparent analog advantage. The claim is retracted, and the surviving char-LM claim is a trade-off rather than a win: **the analog datapath buys a $1.60\times$ proxy-energy reduction for a $+0.160$ bpc (5.3%) quality cost** against a properly regularized reference. Every margin figure in the paper, on both tasks, is stated against that regularized reference. Weight decay applied to the same baseline is worth exactly zero ($+0.0047 \pm 0.0110$ bpc), which dissociates the effect from generic capacity control and pins it to state-level stochasticity in the recurrence — a mechanism the analog datapath supplies physically, and one of the few points that argue *for* analog silicon on grounds other than energy.

And we report a negative on our own theory. Paper 1 (Wang, 2026) derives a firing-floor bound: a recurrent state carried in spikes cannot go below an activity $\rho \geq H_b^{-1}(M \log_2 K/H)$ set by the memory the task demands. We designed a pre-registered test of it — shrink the state width H at fixed memory load, sweeping the predicted floor $12\times$ while keeping the task learnable for the digital reference at every width — and the pre-registered confirmation criterion was refuted. Measured state activity *falls* ($0.496 \to 0.250$) as the predicted floor rises, and spiking-state accuracy *improves* as its own certified capacity shrinks (margin kept $0.079 \to 0.200$). An information bottleneck cannot produce that ordering; an optimization pathology can. We could not construct a regime in which the bound binds on a network that still learns the task, so the bound stays in this paper as theory with a *tested* scope limit, not as validated theory (§7). The empirical rule it was meant to justify — do not carry a compressed recurrent state in spikes — survives, but on the direct evidence of §5 rather than on the bound.

Contributions.

1. A **parameter-matched four-variant protocol** for the neuromorphic-SSM datapath fork: identical tensors, only the nonlinearity's position moves, with emitted-event rate and state activity reported separately so that wire traffic is never confused with state density (§3).

2. **Two-task, matched-communication-rate evidence that the variant ranking inverts** across workloads, at three seeds on both tasks and three memory loads on the recall task (§4, §5).

3. **The datapath-degradation principle**, which predicts both orderings where state fidelity alone

does not, with the quantizer contrast as an out-of-sample (post-hoc identified) test (§6).

4. **Two honest negatives and a co-design constraint**: the firing-floor bound is not empirically validated by this architecture at this scale in either direction (§7); the "analog beats digital" reading is retracted after a regularization control (§4.2); and on an analog datapath the event threshold is floored by the state quantizer's least significant bit (0.125 at 6 bits over ± 4 rails), so event sparsity and state precision are not independent knobs but a converter-area tradeoff (§8).

Scope. Everything here is simulation at small scale — at most 1.7×10^5 parameters, short training schedules, two tasks — and every energy number is a 45 nm operation-count proxy, not a measurement on silicon. The analog storage element and its converter, which are real area and power costs, are not priced by that proxy at all. We state these limits where they bite rather than only in §9, because they bound what the recommendation in §8 can support.

2 Related work

2.1 State-space models

Structured SSMs (Gu et al., 2022b,a) replaced the nonlinear recurrence of an RNN with a linear one whose state transition is diagonal (after reparameterization), which is what makes them both trainable at length and, for our purposes, physically realizable: a diagonal linear recurrence is a bank of independent leaky integrators. We use an S4D-real–style diagonal decay as the shared backbone of all four variants, which is the simplest form that keeps the comparison about the datapath rather than about the recurrence's expressivity.

The design predates the current line. The Legendre Memory Unit (Voelker et al., 2019) derives an optimal linear delay system from a Padé approximant and projects it onto a Legendre basis, arriving at a linear SSM independently of the HiPPO/S4 lineage. It matters here for a reason beyond priority: the LMU was implemented through the Neural Engineering Framework on spiking neuron populations and *deployed on both Loihi and Braindrop* — a digital spiking chip and an analog mixed-signal one — reaching 97.15% on psMNIST against an LSTM's 89.86%. Running an SSM on neuromorphic hardware, including in a non-spiking analog form, is therefore an existence proof rather than a proposal. The open question, and ours, is what each route costs.

2.2 The three datapath routes

(a) Digital SSMs on digital fabric. The reference point: run the exact linear recurrence in a digital datapath, optionally quantized. Our `digital` variant is this route at full precision, and it is the baseline every margin in this paper is measured against.

(b) Continuous state, spiking output. The current "spiking SSM" line keeps the linear state at full precision and applies a spiking nonlinearity only to what leaves the recurrence. SPikE-SSM (Zhong et al., 2024) is the clearest instance: a refractory-LIF nonlinearity on an S4D backbone, with a continuous state h_t. Its reported sparsity figures (around 8% on LRA, 24.5% on WikiText) are therefore *output* sparsity, and its cost is a recall gap (WikiText perplexity 33.2 against S4's 21.0). This distinction is not a technicality; it is the reason a firing-floor argument about state-carrying spikes does not apply to this route at all. Our `spikeout` variant reproduces the design point, and reproduces the signature: its measured recurrent-state activity is exactly 1.0000 in every cell we ran, on both tasks and at every memory load — all of its sparsity is in the output, none of it in the state.

(c) Native analog state. Here the device physics *is* the recurrence. CIM-SSM (Zhang et al., 2026) implements a non-spiking continuous diagonal SSM in a memristor crossbar, using the device's native short-term-memory relaxation as the state decay — the exponential relaxation of the device is e^{At}, so there is no discretization mismatch to manage, and $Ax + Bu$ is an in-memory matrix–vector multiply. Related analog compute-in-memory work approaches the same substrate from adjacent angles: QS4D (Siegel et al., 2025) trains diagonal S4D with quantization-aware training for deployment on tantalum-oxide memristive crossbars and reports that the training itself confers robustness to analog noise. IMSSA (Siegel et al., 2024) is the same group's earlier hardware demonstration: recurrent S4D kernels with A, B and C programmed into a single 64×64 memristive crossbar and executed step-by-step rather than as a convolution, with weights quantized to ternary; on a two-class spoken-digit task the deployed model reaches 81.7% against a 95.1% software reference, a gap the authors attribute to stuck devices — an on-silicon instance of analog imperfection charging quality. HPD (Feng et al., 2025) addresses the robustness side directly, though in simulation rather than on silicon: a layer-wise vulnerability analysis of Mamba under weight perturbations — the noise enters the weights, not the state — locates the damage in the final block's output projection, and the remedy is itself a hybrid datapath split: the projection's SVD factor $U\Sigma$ stays on the analog array while $V^\top$ is offloaded to digital for exact correction, an independent instance of spending exactness on the side of the datapath the loss depends on. LMU-on-Braindrop is the earlier precedent. We treat this route in simulation, with the analog storage element's imperfections modelled explicitly (rails, additive state noise, finite state precision) and communication by send-on-delta graded events; §3 states exactly what is and is not modelled. We do not quote CIM-SSM's reported numbers, which are behind a paywall we could not clear.

2.3 What is missing, and what this paper adds

Two gaps motivate the study.

First, **the routes are never compared under matched capacity**. Each of (a), (b), (c) is developed in its own papers against its own baseline, typically a full-precision model of different size trained under a different schedule, so their efficiency claims are route-internal and not commensurable. Our four variants share every tensor and every hyperparameter; the only difference between them is where a nonlinearity sits.

Second, **energy claims are made at whatever operating point the method happens to reach**. A variant that emits on 27% of steps and one that emits on 65% are not comparable on quality without controlling the rate. We therefore sweep the analog event threshold until the analog variant emits at each spiking comparator's own measured rate, and compare there (§4.1); on the recall task the three non-digital variants land within 0.41–0.50 emission natively, so the comparison is already matched.

A third issue is one we created for ourselves and report because it changes published-looking numbers: **the comparator's regularization**. The lossy analog state acts as a stochastic regularizer at small scale, so an analog variant compared against an untuned digital baseline is credited for a benefit that has nothing to do with its datapath. Giving the baseline the same training-time state noise removes it, and turns an apparent quality win into an energy-for-quality tradeoff (§4.2). We have not seen this control run in the spiking- or analog-SSM literature, and the bias it introduces runs in the same direction everywhere: in favour of the neuromorphic variant.

2.4 Relation to paper 1

Paper 1 (Wang, 2026) established, on a controlled architecture-fixed protocol, that a recurrent network carrying its state in spikes cannot sparsify below roughly 50% activity on a character-LM task, while feed-forward perception sparsifies to 5% and a spiking Transformer to 2% — a firing ceiling specific to *recurrence*. It also derived the bound $\rho \geq H_b^{-1}(\cdot)$ as a candidate explanation. This paper takes up the question that result raises — if spikes are a bad way to carry recurrent state, what datapath *should* carry it? — and, in §7, reports that the bound itself does not survive a direct empirical test at this scale. The empirical ceiling from paper 1 is a measurement and stands; the bound was a prediction about *why*, and it is the prediction that fails.

3 Experimental protocol

3.1 One model, four datapaths

All comparisons in this paper are between four implementations of the *same* state-space model. Every variant owns an identical set of tensors — token embedding, input projection W_{in}, a diagonal decay (an S4D-real–style linear recurrence), a mixing layer W_{mix}, and output projection W_{out} — so parameter counts match exactly (173,596 parameters for the character-LM configuration at H=256; 10,513 to 87,889 for the copy configurations, depending on width). The only thing that moves between variants is *where in the datapath a nonlinearity (spiking or analog degradation) is inserted*:

- **digital** — the continuous baseline: exact state, exact output. This is the reference implementation a GPU or digital accelerator would run.
- **spikeout** — continuous recurrent state, leaky-integrate-and-fire (LIF) nonlinearity on the *output* only, following the SPikE-SSM design point: the linear state is carried at full precision and only the emitted representation is binarized.
- **analog** — the recurrent state is held as a continuous *analog* variable with the imperfections of an analog storage element simulated explicitly: hard rails at ± 4, additive state noise σ=0.02 (applied at training time), and 6-bit uniform quantization over the rail range (applied always, i.e. also at inference). Communication is by *send-on-delta graded events*: a unit transmits only when its quantized state has moved by more than a threshold θ since its last transmission.
- **spikestate** — the recurrent state itself is carried in binary spikes (LIF state). This is the floor-control variant: it is the only datapath inside the scope of the firing-floor bound of Wang (2026), which prices a recurrent memory carried in spikes.

Because the tensors are shared and only the nonlinearity's position moves, any quality or activity difference between variants is attributable to the datapath, not to capacity, architecture, or tuning asymmetries.

3.2 Tasks: one statistical, one precise-recall

The central claim is workload-conditional, so the protocol needs two workloads that isolate the two demands a recurrent state can carry.

Character-level language modeling (statistical). Char-level WikiText, 1.4M characters, vocabulary 65, six epochs, three seeds. The task rewards accurate sequence *statistics*; no individual symbol must be retained exactly. Quality is bits-per-character (bpc).

Synthetic copy (precise recall). The model reads M symbols from an alphabet of K=16, then must reproduce them after a delay; sequence length $L \in \{33, 65, 129\}$ gives memory load $M = (L-1)/2 \in \{16, 32, 64\}$. Chance accuracy is $1/16 = 0.0625$ and chance bpc is $\log_2 16 = 4.000$. Training uses 30 epochs and 80,000 sequences per load, three seeds. This budget was set by an explicit baseline-gate probe: at the original short budget (6 epochs, 20k sequences) even the digital baseline barely cleared chance, making variant comparisons meaningless; at 30 epochs/80k it reaches 9–10× chance at every width and load used here, so all reported cells are legitimate comparisons.

3.3 Reference baseline and controls

A comparison against an under-tuned baseline flatters every alternative. The project learned this the expensive way (Section 4.2), so all margin figures in this paper are referenced to a *noise-regularized* digital baseline: the plain digital model trained with the same additive state noise σ=0.02 the analog datapath receives, applied at training time only. This costs nothing at inference (identical datapath and energy) and is worth −0.309 bpc on char-LM and −0.150/−0.215 bpc on copy at M=16/32 — larger than several of the effects under study, which is exactly why it must be in the reference.

Two dissociation controls pin down what the regularizer is: weight decay (10^{-4}) is worth exactly nothing (+0.005±0.011 bpc), ruling out generic capacity control, and raising the noise dose to σ=0.05 changes nothing (−0.312 vs −0.309), so the dose is already saturated. The benefit is specifically *state-level stochasticity in the recurrence.*

3.4 Metrics

Quality: bpc (char-LM), accuracy and bpc (copy). Variant effects are reported as *paired per-seed* Δbpc against the reference, which removes shared seed variance.

Margin kept: on copy, absolute accuracies are low, so we report the fraction of the reference baseline's *above-chance* accuracy margin a variant retains, $(\text{acc} - 0.0625)/(\text{acc}_{\text{ref}} - 0.0625)$.

Activity, split into two numbers. *Emitted rate* counts transmissions on the wire (spikes, or send-on-delta events); *state activity* counts active state units. The distinction matters: the analog variant's state is dense by construction (state activity ≈ 0.99) while its emitted rate can be low — matching variants on emitted rate matches *wire traffic*, not state density, and a dense analog state costs a storage element and a converter per unit that no spike count captures.

3.5 Energy proxy — and what it is not

Operation counts are priced with the 45 nm figures of Horowitz: 4.6 pJ per multiply-accumulate (MAC) and 0.9 pJ per accumulate (AC). Spiking paths pay AC where a spike replaces a multiply; analog graded events are priced *both* ways (AC-optimistic and MAC-conservative), since a graded event is not a 1-bit spike. We state plainly: these are proxy numbers, not silicon measurements; the analog storage element, its converters, and the ADC are *unpriced*; no claim in this paper should be read as a measured-hardware energy result.

4 Char-LM: the statistical workload

4.1 Results at matched communication rate

Table 1 gives the full three-seed character-LM comparison: the four datapaths, an analog send-on-delta threshold sweep $\theta \in \{0.15, 0.25, 0.5, 0.75, 1.0\}$, and the regularized-baseline arms. Two references are shown, because the choice of reference is itself one of this section's results: paired Δbpc against the *plain* digital baseline, and against the *noise-regularized* digital baseline of Section 3.3.

The obvious objection to any cross-datapath quality comparison is that each variant sits at its own operating point. The θ sweep removes it: the analog variant can be dialed to either spiking comparator's *own* emitted rate and read there.

- **vs. spikestate** (emits 0.645 ± 0.020): analog at θ=0.15 emits 0.612 — slightly *less* wire traffic — at paired Δbpc −0.149 versus spikestate's +0.278: a **0.427 bpc** gap at matched communication.
- **vs. spikeout** (emits 0.369 ± 0.037): analog at θ=0.75 emits 0.380 at Δbpc −0.050 versus +1.055: a **1.10 bpc** gap.

Forcing analog up to the comparators' emission does not erode its advantage; Δbpc is monotone in emission

Table 1: Char-level LM, 3 seeds (mean ± sd), 173,596 parameters per variant, identical budgets. "Emit" is the emitted (wire) rate; "state" is state activity. Paired Δbpc is per-seed against the plain digital baseline (negative = better). The final column re-references against the noise-regularized digital baseline. Energy is the 45 nm proxy of Section 3.5, not a silicon measurement.

Variant	θ	bpc	emit	state	Δbpc (plain)	Δbpc (reg.)	pJ/tok
digital (plain)	–	3.343 ± 0.022	1.000	1.000	0	+0.309	712k
digital + noise reg.	–	3.034 ± 0.017	1.000	1.000	-0.309 ± 0.024	0	712k
analog	0.15	3.194 ± 0.008	0.612	0.994	-0.149 ± 0.014	+0.160	446k
analog	0.25	3.209 ± 0.023	0.550	0.994	-0.134 ± 0.034	+0.175	442k
analog	0.50	3.234 ± 0.003	0.472	0.993	-0.109 ± 0.024	+0.201	438k
analog	0.75	3.293 ± 0.011	0.380	0.992	-0.050 ± 0.020	+0.259	432k
analog	1.00	3.360 ± 0.027	0.266	0.992	$+0.017 \pm 0.007$	+0.326	426k
spikestate	–	3.621 ± 0.013	0.645	0.645	$+0.278 \pm 0.009$	+0.587	157k
spikeout	–	4.398 ± 0.029	0.369	1.000	$+1.055 \pm 0.032$	+1.364	433k

(+0.017 → −0.149 as emission rises 0.27 → 0.61). On the statistical workload, the analog-state datapath is the best neuromorphic route at *every* matched operating point, and the ordering (analog ≪ spikestate ≪ spikeout) is not an operating-point artifact. Note also spikeout's signature, which will matter in Section 5: its state activity is exactly 1.0000 at every seed — output spiking leaves the state fully dense, so its "sparsity" lives entirely on the wire.

4.2 The control that changed the headline

Read against the plain digital baseline, Table 1 appears to show the analog datapath *beating* its digital reference by 0.149 ± 0.014 bpc at 61% emission — all three seeds negative. We report this intermediate finding because we published it and then retracted it, and the retraction is the section's real result.

The suspicion is visible in the mechanism: the analog state's noise, quantization, and send-on-delta suppression act as a stochastic regularizer on a 173k-parameter model at a 6-epoch schedule, and the plain digital baseline is not regularization-tuned. The control is to give the baseline the same medicine. Training-time state noise σ=0.02 on the digital baseline is worth -0.309 ± 0.024 bpc — *2.1 times the entire apparent analog advantage* — while weight decay is worth exactly nothing ($+0.005 \pm 0.011$) and 6-bit quantization contributes nothing on this task (−0.311 with vs. −0.309 without). Against the properly regularized baseline, analog at θ=0.15 is +0.160 bpc *worse*, far outside seed noise.

The clean decomposition of the analog datapath's net effect on char-LM:

$$\underbrace{-0.309}_{\text{state-noise gift}} + \underbrace{+0.160 \text{ to } +0.326}_{\substack{\text{send-on-delta gating cost} \\ \theta=0.15\to\theta=1.0}}$$

The analog datapath supplies a genuine regularizer for free — and physically, which is a real point in favor of analog silicon — but its sparsity mechanism charges more than the gift is worth at every threshold.

4.3 The surviving claim

Because the reference's noise is training-time only, the regularized digital baseline has an *identical inference datapath and energy* to the plain one (712k pJ/token proxy). The honest char-LM headline is therefore an energy–quality tradeoff, not a quality win:

> **On the statistical workload, the analog-state SSM buys a 1.60× proxy-energy reduction (446k vs. 712k pJ/token) for a +0.160 bpc (5.3%) quality cost against a noise-regularized digital baseline, while remaining — by 0.43 and 1.10 bpc at matched wire rates — the best of the three neuromorphic datapaths.**

Re-referencing shifts every variant by the same constant, so the variant ordering is unaffected: against the regularized baseline, analog +0.160 ≪ spikestate +0.587 ≪ spikeout +1.364. Two caveats travel with the headline: the regularization-mediated effects here may shrink at larger scale or under a fully tuned baseline, and matching emitted rate matches wire traffic only — the analog state is dense (state activity 0.994), and its storage elements and converters are unpriced by the proxy (Section 3.5).

5 Copy: the precise-recall workload

The copy task inverts every conclusion of Section 4, and the inversion is the pivot of this paper. The model must reproduce $M \in \{16, 32, 64\}$ symbols from a K=16 alphabet after a delay (chance accuracy 0.0625, chance bpc 4.000); unlike character-level language modeling, success requires *precise retention* of specific symbols rather than sequence statistics.

5.1 Results and the ranking inversion

Table 2 gives the three-seed comparison at all three memory loads. "Margin kept" is the fraction of the reference baseline's above-chance accuracy margin a variant retains — the appropriate metric when absolute accuracies are low. Following Section 3.3, the reference at $M{=}16$ and $M{=}32$ is the *noise-regularized* digital baseline, which is stronger than the plain one on copy as well (paired Δbpc -0.150 ± 0.033 at $M{=}16$, -0.215 ± 0.011 at $M{=}32$); margins against the plain baseline are 3–5 points higher and are upper bounds.

The three non-digital variants emit at nearly matched rates (0.40–0.50 at every load), so — as in Section 4.1 — the quality spread is not an operating-point artifact. And the ordering is the exact reverse of char-LM: at $M{=}16$, best to worst is **spikeout** (margin kept 0.87), **analog** (0.50), **spikestate** (0.074); on char-LM the same three variants ranked analog (Δbpc +0.160 vs. the regularized reference) $\ll$ spikestate (+0.587) $\ll$ spikeout (+1.364). The variant that was worst on the statistical workload — continuous state, spiked output — is the only one that survives precise recall, keeping 87% of the regularized baseline's margin at a 45% emission rate and 3.2× lower proxy energy.

5.2 A one-bit recurrent state eliminates recall

The spikestate row is not merely degraded. At $M{=}32$ it lands at *exactly* chance — accuracy 0.0623 ± 0.0012 against 0.0625, bpc 4.005 ± 0.004 against 4.000 — and reproduces this at $M{=}64$ (0.0623 ± 0.0005). A one-bit recurrent state does not degrade precise-recall performance at these loads; it *eliminates* it. It is simultaneously the cheapest cell in the table (102k pJ/token at $M{=}32$, 4× below the digital baseline) and buys nothing: the cleanest demonstration in this study that an energy number is meaningless without the task quality it purchased. For memory-bearing workloads, carrying the recurrent state in spikes is a non-option, not a cheap option.

5.3 Load dependence, and the validity limit at $M{=}64$

Between $M{=}16$ and $M{=}32$ — the loads where the digital reference still learns the task well above chance — the quality gap *widens* for both surviving neuromorphic routes: margin kept falls $0.87 \to 0.42$ for spikeout and $0.50 \to 0.28$ for analog (seed sd ≤ 0.003 acc, so far outside noise). The analog variant's emission does not rise to compensate at fixed θ; it *falls* ($0.501 \to 0.449$). Two consequences: there is no single "X% of digital" figure for a neuromorphic SSM — the workload's retention demand sets it — and any analog energy/quality curve must be drawn per memory load.

The $M{=}64$ column is reported for completeness but is *validity-limited*, and we do not use it for claims. The digital baseline itself is close to floor there (accuracy 0.167, only 2.7× chance, against 10× chance at $M{=}16$), so every variant compresses toward chance and both metrics lose resolution: margin kept rebounds non-monotonically (spikeout $0.92 \to 0.46 \to 0.67$ vs. the plain baseline) and paired Δbpc compresses ($+0.71 \to +0.16$ for spikeout from $M{=}32$ to $M{=}64$) — a shrinking-reference artifact, not a recovery. The regularization control was also not run at $M{=}64$, so no corrected margin exists there. The load-dependence claim above is therefore made on $M{=}16 \to 32$ only.

6 The datapath-degradation principle

6.1 Statement

Sections 4 and 5 give matched-communication-rate variant orderings on two workloads, and they invert. One principle predicts both:

> *A neuromorphic SSM datapath is cheap exactly when it degrades the part of the computation the workload does not depend on.* Statistical sequence workloads need an exact graded *output* distribution and tolerate a lossy state, so the analog-state route is the best neuromorphic option there. Precise-recall workloads need an exact recurrent *state* and tolerate a spiked output, so the output-spiking route is the only neuromorphic option that survives there.

6.2 Why state fidelity alone is not the axis

A simpler candidate explanation — rank the variants by how exact their recurrent state is — fits the copy ordering perfectly: at matched emission, margin kept falls monotonically with state fidelity, continuous (0.87) > lossy-analog (0.50) > one-bit (0.074). But it fails on char-LM: the output-spiking variant has the *most* exact recurrent state on both tasks (state activity exactly 1.0000 ± 0.0000 in every cell of both tables, the most reproducible measurement in this study) yet is the *worst* variant on char-LM and the best on copy. State fidelity predicts one table; only the two-sided principle — which part of the datapath does the task read? — predicts both.

Table 2: Copy task, 3 seeds (mean ± sd), ep30 budget, H=256 (87,889 parameters), K=16, chance acc 0.0625 / bpc 4.000. "Emit" is the emitted (wire) rate; "state" is state activity. Margin kept is vs. the noise-regularized digital reference at M=16/32 and vs. the plain digital baseline at M=64 (the regularization control was not run there; see §5.3). Energy is the 45 nm proxy, not a silicon measurement.

M	Variant	acc	bpc	emit	state	margin kept	pJ/tok
16	digital (plain)	0.630 ± 0.011	1.420 ± 0.034	1.000	1.000	0.95	398k
16	digital + noise reg.	0.663 ± 0.009	1.270 ± 0.029	1.000	1.000	1.00	398k
16	spikeout	0.583 ± 0.012	1.545 ± 0.071	0.447	1.000	0.87	123k
16	analog θ=0.2	0.361 ± 0.051	2.777 ± 0.226	0.501	0.972	0.50	247k
16	spikestate	0.107 ± 0.016	3.929 ± 0.037	0.496	0.496	0.074	108k
32	digital (plain)	0.315 ± 0.003	2.893 ± 0.013	1.000	1.000	0.90	398k
32	digital + noise reg.	0.343 ± 0.003	2.678 ± 0.016	1.000	1.000	1.00	398k
32	spikeout	0.180 ± 0.001	3.605 ± 0.003	0.485	1.000	0.42	125k
32	analog θ=0.2	0.142 ± 0.000	3.788 ± 0.002	0.449	0.964	0.28	231k
32	spikestate	0.062 ± 0.001	4.005 ± 0.004	0.405	0.405	0.00	102k
64	digital (plain)	0.167 ± 0.001	3.666 ± 0.012	1.000	1.000	1.00	398k
64	spikeout	0.133 ± 0.001	3.824 ± 0.002	0.481	1.000	0.67	125k
64	analog θ=0.2	0.110 ± 0.001	3.914 ± 0.001	0.401	0.955	0.46	216k
64	spikestate	0.062 ± 0.001	4.002 ± 0.001	0.238	0.238	0.00	91k

6.3 An out-of-sample test: the quantizer contrast

The principle makes a falsifiable prediction about any single degradation applied to the *same* datapath on both workloads: an inference-time loss of state precision should be near-free on the statistical task and costly on the precise-recall task. The regularization control of Section 3.3 contains exactly such a degradation: the 6-bit state quantizer (unlike the additive noise, which is training-time only, the quantizer acts at inference). Applied to the *digital* baseline at matched noise σ=0.02, the same quantizer costs:

Workload	Δbpc	state footprint
char-LM	−0.002	—
copy M=16	+0.369	1.000→0.971±0.015
copy M=32	+0.385	1.000→0.976±0.005

Both copy rows are three seeds, and Δbpc is the quantizer-alone cost at matched σ=0.02. The identical hardware degradation is free on the statistical workload and costs $\sim$ 0.37–0.39 bpc on the precise-recall workload — the principle's prediction, out of sample, on a datapath (plain digital) that none of the four variants' comparisons touched. The effect is load-robust: a floor of "$\geq$ +0.37 at M=32, smaller counts as evidence against" was pre-registered after the M=16 result and met (+0.385 vs. +0.369; flat to slightly rising, within arm sd $\sim$ 0.04–0.05 — we say "does not attenuate with load," not "grows"). Quantization alone still keeps 0.78 of the digital margin at M=32, far above spikeout's 0.42 and analog's 0.28, so state-precision loss explains part but not most of the neuromorphic copy deficit; the event and spike mechanisms add their own cost.

Honesty label: this test is post-hoc-identified. The control row was pre-registered with the *noise* arm bound to the principle's prediction, and that assignment was wrong: inspection of the implementation showed the additive noise is gated behind training (`if self.training`) while the quantizer is not, so the noise arm is a training-time regularizer that tests baseline tuning, not an inference datapath degradation — and its landing (noise *helps* on copy too) says nothing about the principle. The quantizer arm is the row's only inference-time state degradation, and it was identified as the real test only after the arms landed. We therefore report the contrast as post-hoc but mechanistically motivated, with one genuinely pre-registered element (the M=32 floor, set before those cells finished). The pre-registered test *as written* failed by mis-specification.

6.4 What weakens the principle

Two observations run against a strong reading. First, the training-time noise benefit does not separate the workloads the way a naive reading would hope: it is -0.150 ± 0.033 bpc at copy M=16, -0.215 ± 0.011 at M=32, and -0.309 ± 0.024 on char-LM — the trend across loads runs *toward* the char-LM value, so "the regularizer helps the statistical task more" is at best a weak supporting line and we do not lean on it. Second, the principle has been tested out of sample exactly once (the quantizer contrast), on two tasks and one architecture family; Section 9 states the scale caveats.

7 The firing-floor bound: theory with a tested scope limit

Paper 1 (Wang, 2026) derives a firing-floor bound for recurrent networks that carry their state in spikes: if a binary state of width H with mean activity ρ must distinguish the memory states a task demands, then

$$\rho \;\geq\; H_b^{-1}\left(\frac{M \log_2 K}{H}\right), \qquad (1)$$

where H_b is the binary entropy function. The numerator counts *bits*, not symbols: copying M symbols drawn from an alphabet of size K requires distinguishing K^M states, so the demand is $M \log_2 K$ bits.[1] The bound is the theoretical reason to expect a spiking recurrent state to be dense, and it motivated the whole three-route comparison. This section reports what happened when we tried to make it bind on a network that still learns its task. It does not bind, in either direction, and we report that as the negative it is.

7.1 Raising the memory load cannot test it

The obvious test is to raise M until the floor is high enough to measure against. It fails structurally, not for want of compute. On the copy task the `spikestate` variant is at exactly chance by M=32 (accuracy 0.0623 ± 0.0012 against a chance level of 0.0625, three seeds) and again at M=64 (0.0623 ± 0.0005). At H=256 the floor only reaches 0.500 at M=64 — a load at which the network retains nothing. Equation (1) prices a state that actually carries the sequence, so a network storing nothing is outside its scope regardless of what activity it happens to exhibit. *The learnability ceiling of a spiking recurrent state on copy ($M \approx 16$) sits below the load at which the floor begins to bind (M=64).* Consistent with this, measured state activity *falls* with load (0.496 ± 0.011 at M=16, 0.405 ± 0.030 at M=32) rather than rising as the bound would require — but we do not read that as evidence against the bound, because the premise of retention is already violated. Copy columns must never be cited as bound tests.

7.2 Shrinking the width is the valid design

The alternative moves the same ratio $M \log_2 K/H$ by shrinking the denominator. Holding M=16 (the load a spiking state can still partly learn, K=16, so a fixed 64-bit demand) and sweeping $H \in \{256, 128, 96, 64\}$ moves the predicted floor $0.042 \to 0.110 \to 0.174 \to 0.500$, a 12× swing, on a task whose difficulty is unchanged. Both arms were run at three seeds per cell, 30 epochs, 80k copy sequences (Table 3).

Four criteria were registered in the results ledger before any cell of this row landed: (i) a *validity gate* — the test counts only at widths where the digital reference still learns copy well above chance; (ii) *confirmation* — `spikestate`'s state activity sits at or above the predicted floor *and rises as H shrinks*, the trend being the discriminating observable since the H=256 point already sits far above its own floor; (iii) *refutation* — activity stays flat or falls across the sweep while the networks still learn, which would make the observed $\rho \approx 0.25$–0.5 an optimization artifact of the LIF state rather than an information-theoretic floor; (iv) a stated *most likely outcome* — retention may die before the floor binds, in which case the honest verdict is that the bound is not empirically testable with this architecture at this scale.

The validity gate passes everywhere. The digital reference keeps 9–10× chance accuracy across an 8.4× parameter range ($0.630 \to 0.560$ from H=256 to H=64). Copy at M=16 is therefore not width-limited anywhere in $H \in [64, 256]$: shrinking H moves the predicted floor by 12× while barely changing what the task demands of a network that can do it. That is precisely the property raising M could not deliver, and it makes all four cells legitimate tests rather than capacity-limited ones.

Criterion (ii) is refuted, by anti-correlation rather than by flatness. As the predicted floor rises 12×, measured state activity *falls* by half: $0.496 \to 0.392 \to 0.391 \to 0.250$. The H=128 and H=96 points are statistically indistinguishable, and at H=64 the measured activity lands *below* its own predicted floor. This is criterion (iii)'s shape. Measured LIF state activity in this architecture is **width-determined, not information-determined**.

7.3 The anti-capacity finding

The sweep also produces an ordering that no information bottleneck can generate. Inverting (1) turns a measured (ρ, H) into a capacity certificate $H \cdot H_b(\rho)$: the most bits the spiking state could be carrying. Across the row that certificate falls $256.0 \to 123.6 \to 92.7 \to 51.9$ bits against a fixed 64-bit demand, i.e. from a 4.0× surplus to a 0.81× deficit. *Accuracy moves the other way*: margin kept rises monotonically $0.079 \to 0.169 \to 0.179 \to 0.200$, and paired Δbpc against the digital reference improves in lockstep ($+2.509 \pm 0.060 \to +2.079 \pm 0.016 \to +2.050 \pm 0.023 \to +1.829 \pm 0.051$). The most over-provisioned network is the worst one. A capacity constraint cannot produce that ordering; an

[1] An earlier reading of this project used $\log_2 M$, which would have made every cell below unmeasurable (predicted floors $\sim 10^{-3}$). All floors quoted here use the corrected form.

Table 3: Shrink-H bound row. Copy at L=33 (M=16, K=16, i.e. a 64-bit memory demand), 30 epochs, 80k sequences, three seeds per cell; chance accuracy 0.0625. Floor is $H_b^{-1}(64/H)$. ρ is measured `spikestate` recurrent-state activity; margin kept is the fraction of the digital reference's above-chance accuracy margin that `spikestate` retains; the certificate is $H \cdot H_b(\rho)$ bits, to be compared against the fixed 64-bit demand.

H	params	floor	digital acc	spikestate acc	ρ	margin kept	certificate
256	87,889	0.042	0.6302 ± 0.0106	0.1071 ± 0.0164	0.4960 ± 0.0106	0.079	256.0
128	28,113	0.110	0.5944 ± 0.0023	0.1522 ± 0.0049	0.3915 ± 0.0417	0.169	123.6
96	18,289	0.174	0.5889 ± 0.0065	0.1570 ± 0.0063	0.3909 ± 0.0336	0.179	92.7
64	10,513	0.500	0.5600 ± 0.0095	0.1619 ± 0.0039	0.2496 ± 0.0252	0.200	51.9

optimization pathology can. Spiking-state failure on precise recall is a trainability failure, and the bound does not explain it.

A retracted intermediate reading, kept for the record. When only the H=64 cell had landed, its 12-bit certificate deficit alongside a large accuracy shortfall looked like the bound earning its keep in contrapositive form — predicted incapacity, observed failure. The H=96 cell killed that reading: the certificate there shows a 1.4× *surplus* and `spikestate` nevertheless fails essentially identically (0.157 vs. 0.162 accuracy). The match at H=64 was coincidental, and the certificate is necessary-but-not-sufficient and non-diagnostic at this scale. We state this because the intermediate reading was recorded before it was refuted, and because the refutation is the more informative result.

7.4 Verdict

Equation (1) is **not empirically validated by this architecture at this scale, in either direction**. We could not construct a regime where the firing-floor bound binds on a network that still learns the task: raising the memory load kills retention before the floor engages, and shrinking the width engages the floor but finds measured activity anti-correlated with it across a 12× swing, with accuracy improving as predicted capacity shrinks. The bound stays in this paper as theory with a stated and now *tested* scope limit — it applies to a spiking recurrent state that genuinely carries the task's memory, and we were unable to build one.

Two things survive this negative and are worth separating from it. The empirical firing ceiling of Paper 1 — a recurrent spiking char-LM that will not sparsify below ≈ 50% — is a measurement, not a prediction of (1), and is untouched. And the practical design conclusion that a compressed recurrent state should not be carried in spikes is established here empirically rather than theoretically: Section 5 shows a 1-bit recurrent state sitting at exactly chance on precise recall at two memory loads while being the cheapest variant by the energy proxy. The bound was the reason to expect that; it is not the reason to believe it.

8 Hardware implications

The preceding sections were organized around a scientific question — which datapath degradations does a workload tolerate? This section reads the same results as design guidance: which route should carry a state-space model on neuromorphic silicon, and what co-design constraints bind once a route is chosen.

8.1 The recommendation splits by workload

No route wins both tasks, and the ordering inversion of Section 5 is stable at three seeds on both sides, so the recommendation cannot be collapsed to a single answer.

Statistical sequence workloads with low retention demand — perception streams, sensor fusion, language-like statistics — tolerate a lossy recurrent state, and there the analog/compute-in-memory route is the fit: on char-LM it holds the state in analog dynamics, communicates on 61% of steps, and pays +0.160 bpc (5.3%) against a noise-regularized digital baseline for a 1.60× proxy-energy reduction (Section 4.3). This is a trade-off, not a free lunch, but it is the only cell in either task where a neuromorphic datapath buys energy at single-digit quality cost.

Precise-recall workloads invert the choice. On copy the analog route loses to output-spiking on *both* axes (Section 8.3), and the route that survives is the one SPikE-SSM instantiates: keep the state in an exact digital datapath and spike only the output, which retains 0.87 and 0.42 of the regularized baseline's above-chance margin at M=16 and M=32 while emitting on ∼45% of steps.

Carrying the recurrent state in spikes is a non-option for any memory-bearing workload. At M=32 and again at M=64 the spiking-state variant sits at exactly chance (0.0623±0.0012 and 0.0623±0.0005 against 0.0625) — a 1-bit state does not degrade precise recall, it eliminates it (Section 5.2). The live design fork on real silicon is therefore binary: an exact digital state with a spiked output, or an analog state that accepts the loss and gains the event sparsity. Which side of the fork is right is set by the workload's retention demand, not by the hardware.

8.2 Event sparsity and state precision are not independent knobs

The analog route's operating point is set by the send-on-delta threshold θ, and the natural assumption is that θ is a free knob: lower it for quality, raise it for sparsity. Two measured constraints break that assumption.

First, *the ADC step floors the event rate.* The simulated analog state is quantized to b=6 bits over rails $\pm r$ with r=4, giving a least significant bit of $q = 2r/2^b = 0.125$, and the send-on-delta comparison is made on the *quantized* state. Every θ below q is therefore inoperative: θ=0.05 and θ=0.10 produce bit-identical results on every metric to all recorded digits. Below one LSB the quantizer, not the threshold, decides which state changes are visible, so the minimum achievable event rate is set by converter precision. Buying a lower event rate means spending bits — a converter area and energy cost — not tuning a threshold. This couples two quantities a designer would prefer to optimize separately, and it is invisible in any formulation that treats the state as continuous.

Second, θ *does not transfer across workloads.* The char-LM-calibrated θ=1.0 collapses the analog variant to chance on copy, emitting on only 3–8% of steps; the usable window on copy is $0.125 < \theta \lesssim 0.3$, and θ=0.5 already over-gates to chance. The window's lower edge is the LSB constraint above; its upper edge is task-dependent. An analog SSM deployment therefore needs per-workload threshold calibration, and a fixed-θ part cannot serve both workload classes of Section 8.1.

8.3 What the energy proxy does and does not license

All energy figures in this paper are the 45nm proxy of Section 3.5 (4.6 pJ per MAC, 0.9 pJ per AC); none are silicon measurements. Read under that caveat, the proxy licenses one claim and blocks two others.

It licenses the char-LM tradeoff: analog at θ=0.15 costs 446k pJ/token against the regularized digital baseline's 712k — a 1.60× reduction, purchased for +0.160 bpc. The digital baseline's noise regularizer is training-time only, so its inference datapath and energy are identical to the plain baseline's; the comparison is fair on both axes.

It blocks any analog energy claim on copy. At M=16 the output-spiking variant keeps 0.87 of the regularized margin at 122.9k pJ/token (3.2× below digital's 398.0k), while analog keeps 0.50 at 246.7k — worse on quality *and* energy; at M=32 the pattern repeats (0.42 at 125.2k versus 0.28 at 231.0k). The datapath reason is structural: the analog variant earns event pricing only on the recurrent mixing matrix, while its input and output projections remain MAC-priced, so its proxy energy cannot approach output-spiking's unless most parameters sit in the recurrence.

And it blocks reading cheapness as merit. The cheapest cell in the entire copy table is the spiking-state variant at M=32 (102k pJ/token) — the same cell that retains exactly nothing. A pJ/token figure is meaningless without the quality it purchased.

Finally, the proxy itself omits costs that fall specifically on the analog route: the analog state is dense (state density 0.994), and each unit carries an analog storage element and a converter that the MAC/AC accounting does not price. Matching variants on emitted rate matches wire traffic, not state-holding cost (Section 3.4). The honest summary for a designer is that the analog route's advantage is measured here in communication and quality, and its storage-side costs remain to be priced on silicon — which is why measured Loihi-2/SpiNNaker-2-class energy, not more simulation, is the natural next step.

9 Limitations

Simulation, not silicon. Every energy figure in this paper is the 45 nm Horowitz MAC/AC proxy applied to counted operations, not a measurement of any device. The proxy omits costs that fall specifically on the analog route: the analog state is dense (state density 0.994), and each hidden unit implies an analog storage element and a converter that MAC/AC accounting does not price. Matching variants on *emitted* rate therefore matches wire traffic, not state-holding cost (Section 3.4). The 1.60× figure of Section 4.3 is a claim about communication and arithmetic under the proxy, and could shrink or invert once storage and conversion are priced on real silicon; measured Loihi-2/SpiNNaker-2-class energy is the direct next step.

Small scale, short schedules. The char-LM models have 173,596 parameters trained for six epochs; the copy models span 10.5k–87.9k parameters at thirty epochs. Effects that are regularization-mediated at this scale may attenuate at larger scale or longer schedules. This cuts in both directions: the training-time noise benefit on the digital baseline (−0.309 bpc, Section 4.2) is exactly the kind of small-model, short-schedule effect that typically shrinks as models grow — but the analog route's +0.160 bpc gating cost is measured against that same baseline, so the surviving tradeoff could move in either direction at scale. Absolute task quality is far from state of the art by construction (a one-layer diagonal SSM), which is the price of the parameter-matched protocol.

Two tasks, one out-of-sample test. The ranking inversion rests on one statistical task (char-level WikiText) and one precise-recall task (synthetic copy). The datapath-degradation principle was tested out of sample exactly once — the quantizer contrast of Section 6.3, -0.002 bpc on char-LM versus $+0.369/+0.385$ bpc on copy at $M=16/32$ — and that test is post hoc: the pre-registered arm turned out to be a training-time-only regularizer and could not test the principle as designed. The genuinely pre-registered element is the $M=32$ floor ($\geq +0.37$ bpc, met). A designed, pre-registered test on a third workload family (e.g. associative recall with distractors, or a streaming perception task) is the obvious strengthening experiment.

The regularized reference is not an exhaustively tuned one. All corrected copy margins are quoted against a digital baseline given training-time state noise at $\sigma=0.02$ — the dose at which the char-LM benefit saturated — not against a baseline tuned over a broad regularizer grid. Weight decay was worth exactly zero ($+0.005 \pm 0.011$ bpc), which narrows the mechanism but does not exhaust the space; a stronger baseline would shrink every neuromorphic margin further, and no result in this paper moves in the neuromorphic routes' favour if the baseline improves.

Scope of the bound-arm negative. Section 7 shows the firing-floor bound's width prediction fails on *this* architecture at *this* scale — a one-layer LIF-state SSM whose copy failure is an optimization pathology. That is a tested scope limit, not a general refutation: an architecture that actually trains to retain $M \log_2 K$ bits in a spiking state could still exhibit the floor. We could not construct one, and report that fact rather than the bound's failure in general.

Validity limit at the heaviest load. The copy $M=64$ column is capacity-limited for every variant (the digital reference itself is only 2.7× chance) and has no regularized reference, so it supports only the chance-level reproduction of the spiking-state result; the load-scaling claims of Section 5.3 are scoped to $M=16 \to 32$.

10 Conclusion

The question this paper set out to answer — which datapath should carry a state-space model on neuromorphic hardware — turns out not to have a single answer, and the shape of the non-answer is the contribution. Under a parameter-matched protocol where only the position of the nonlinearity moves, the three neuromorphic routes reverse their ranking between a statistical sequence task and a precise-recall task, at matched communication rates and three seeds on both sides. The regularity behind the reversal is the datapath-degradation principle: a route is cheap exactly when it degrades the part of the computation the workload does not depend on. Statistical workloads need an exact graded output and tolerate a lossy analog state, where the analog route buys a 1.60× proxy-energy reduction for a $+0.160$ bpc (5.3%) quality cost against a properly regularized digital baseline. Precise-recall workloads need an exact state and tolerate a spiked output — and carrying the state itself in spikes is not a degraded option there but a non-option, landing at exactly chance at $M=32$ and $M=64$ while being the cheapest cell in the table.

For a designer, the principle converts into a rule that can be applied before any simulation: identify which side of the datapath — recurrent state or emitted output — the workload's loss actually depends on, and spend the exactness budget there. The co-design constraints that come with the analog route are equally concrete: the ADC step, not the send-on-delta threshold, floors the event rate, so event sparsity is purchased with converter bits rather than a free algorithmic knob, and the threshold does not transfer across workloads.

The firing-floor bound that motivated this study ends in a different place than intended. Across a 12× swing in its predicted floor, measured spiking-state activity moved the other way, and accuracy improved as certified capacity shrank; the bound is not empirically validated in either direction at this scale. It survives as design intuition — the reason to *expect* that a compressed state should not be carried in spikes — while the rule itself now rests on the direct measurements of Section 5: the bound was the reason to expect that result; it is not the reason to believe it. What would move this from simulation to engineering is measured energy on real neuromorphic silicon, with the analog state's storage and conversion priced — the one number this paper cannot supply and the next one should.

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